A CONSTRAINT-AWARE DECISION SUPPORT MODEL FOR PROGRAM AND COLLEGE SELECTION

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Chapter 1

Introduction

Academic routing and college program selection represent a highly complex, multi-criteria decision-making (MCDM) challenge for graduating high school students. In educational institutions, this process is traditionally handled through manual guidance counseling or simple linear digital forms. While these methods serve a basic administrative function, they lack the computational integrity required to evaluate complex, multi-layered student data against strict institutional prerequisites. From a Computer Science perspective, this matching process is fundamentally a "state-transition" and constraint satisfaction problem.

Modern decision support systems must not only store data but actively compute validity. When evaluating student profiles against academic programs, the logic must navigate both non-negotiable hard constraints (e.g., mandatory prerequisites) and subjective soft preferences (e.g., vocational interests and skills). The absence of formal computational models in existing local systems leads to procedural bypasses, algorithmic bias, and illogical program recommendations. Therefore, this study proposes a Constraint-Aware Decision Support Model, utilizing Boolean logic, the Simple Additive Weighting (SAW) algorithm, and Explainable Artificial Intelligence (XAI). In adherence to BS Computer Science (BSCS) standards, a Progressive Web Application (PWA) will be developed specifically to implement and computationally validate the proposed decision model.

Background of the Study

The transition from Senior High School (SHS) to higher education in the Philippines is governed by strict academic alignments. The Department of Education (DepEd) mandates that

SHS strands must align with chosen college degree programs to ensure academic readiness. However, the manual enforcement of these policies is currently failing. Descriptive research by Caballes et al. (2022) revealed an alarming 39% strand mismatch rate among enrolling college students. This misalignment is mathematically detrimental; Punongbayan and Arapoc (2025) proved through propensity score matching that students who mismatch into science and engineering programs without a STEM background suffer a severe, quantifiable "grade penalty."

The computational limitation of current guidance systems—whether physical or digitized via CRUD (Create, Read, Update, Delete) databases—is their inability to execute active dependency resolution. They function merely as passive data repositories that rely on human intervention to verify if a student meets the GWA, strand, or tuition requirements of a program. Furthermore, while some institutions have attempted to automate recommendations using additive algorithms, these standalone models are computationally flawed when handling strict prerequisites. Additive algorithms operate on a mathematical "union," allowing high scores in one area to dangerously mask the failure of a fundamental requirement (Zheng et al., 2023).

Additionally, as the global tech industry shifts toward algorithmic transparency (Sharma et al., 2024), utilizing "black-box" recommendation engines in education is no longer ethically or practically viable. Students and counselors must understand why a specific program was mathematically recommended without the system hallucinating fake human empathy (Graff, 2025).

To resolve these computational limitations, there is a critical need for a hybrid architecture. This study develops a model that first applies Boolean intersection logic to filter hard constraints, subsequently applies the SAW algorithm against standardized O*NET taxonomies for preference ranking, and translates the computational weights into causal text via

XAI. The development of a client-side PWA serves as the necessary testing environment to validate the correctness, consistency, and efficiency of this integrated computational approach.

Statement of the Problem

Selecting a college program is a multi-criteria decision-making (MCDM) task that requires balancing strict relational constraints with dynamic, standardized profiling. While existing systems utilize basic database filtering, bridging the gap between raw occupational data and constraint satisfaction requires a robust computational architecture. From a Computer Science perspective, this requires a two-phase mathematical model capable of absolute elimination through Boolean logic, followed by the algorithmic ranking of negotiable variables.

To address this gap, this study aims to design, develop, and evaluate a Constraint-Aware Decision Support Model for college program selection utilizing relational database querying, the Simple Additive Weighting (SAW) algorithm, and O*NET data integration.

Specifically, this study seeks to answer the following questions:

1. How can relational data architecture and Boolean logic be mathematically modeled to eliminate invalid options based on non-negotiable hard constraints (academic Strand, GWA, and Tuition)?
2. How can the Simple Additive Weighting (SAW) algorithm be implemented to normalize and rank feasible programs, prioritizing standardized O*NET mapping (Skills and Interests) over secondary soft constraints (Location and Institutional Type)?

3. How can Explainable AI (XAI) principles be integrated to dynamically translate the computational weights and Boolean elimination logic into transparent, human-readable justifications?
4. What is the computational performance of the developed model in terms of rule execution correctness, algorithmic consistency, and data retrieval efficiency?

Theoretical Framework

This study is anchored on the intersection of computational optimization, decision mathematics, and occupational taxonomy to formulate a robust academic routing engine. The proposed Constraint-Aware Decision Support Model derives its underlying logic from four foundational theories: Constraint Satisfaction Theory, Multi-Criteria Decision Making (MCDM), Holland's Typological Theory mapped through the O*NET Framework, and Causal Explainable Artificial Intelligence (XAI). Together, these theories dictate the structural flow and algorithmic boundaries of the developed system.

The primary computational foundation of the model is anchored on the principles of Probability-Based Multi-Objective Optimization and Constraint Satisfaction, as articulated by Zheng et al. (2023). From the viewpoint of set theory, traditional additive algorithms operate on a logic of "union," which inherently allows an extreme positive value in one variable to mathematically compensate for the failure of a critical prerequisite. To resolve this flaw in environments with strict relational rules, decision logic must incorporate "intersection" properties. This theory dictates the first phase of the proposed system: before any preference scoring occurs, mandatory criteria (such as GWA, Tuition Budget, and SHS Strand) must act as definitive Boolean boundaries. By treating these variables as hard constraints, the system ensures

that structurally invalid academic programs are instantly eliminated, preventing mathematical compensation from masking fundamental prerequisites.

Once the dataset of viable programs is established through constraint satisfaction, the study relies on the theoretical mechanics of Multi-Criteria Decision Making (MCDM), specifically the Simple Additive Weighting (SAW) method. As demonstrated by Pratiwi et al. (2025) and Nainggolan and Simangunsong (2025), SAW is highly effective in computational decision support systems due to its capacity to execute clear, weighted linear calculations across multiple conflicting criteria. This theory underpins the second phase of the proposed model, dictating how the system normalizes both beneficial and unbeneficial variables and applies mathematical weights to rank the surviving alternatives with high algorithmic transparency.

To ensure the data fed into the SAW algorithm is structurally valid, the study integrates Holland's Typological Theory of vocational environments, mapped computationally through the standardized ONET Framework. Cruz Eraso et al. (2016) established that computational tools for vocational guidance must rely on rigid psychological taxonomies to accurately predict person-environment fit. Building upon this, Hanna et al. (2026) demonstrated that extracting specific occupational task statements from ONET provides significantly higher predictive validity than using abstract interest scales. This taxonomic theory justifies the system's reliance on the 65-item O*NET assessment, ensuring that the subjective inputs calculated by the SAW algorithm are rooted in standardized, empirically validated metrics rather than ambiguous student text.

Finally, the study integrates the theoretical framework of Causal Explainable Artificial Intelligence (XAI) as defined by Graff (2025). As computational models grow more complex, ethical computing mandates that their decisions remain interpretable to end-users. However,

Graff theorizes that attempting to make black-box models provide "reason explanations"—which mimic human empathy or motivations—creates dangerous user over-reliance and false expectations. Instead, algorithmic transparency must be strictly causal. This theory governs the final phase of the proposed system, dictating that the XAI module must not hallucinate psychological advice, but rather strictly translate the mathematical weights of the SAW algorithm and the Boolean logic of the hard constraints into transparent, causal text justifications.

Conceptual Framework

The conceptual framework of this study operationalizes the established theoretical principles into a tangible computational pipeline. To visualize the logical relationship between the user data, the algorithmic execution, and the final system generation, this study utilizes an Input-Process-Output (IPO) architecture. This model maps the deterministic flow of data through the Constraint-Aware Decision Support Model, illustrating how raw student profiles are computationally transformed into explainable academic routing recommendations.

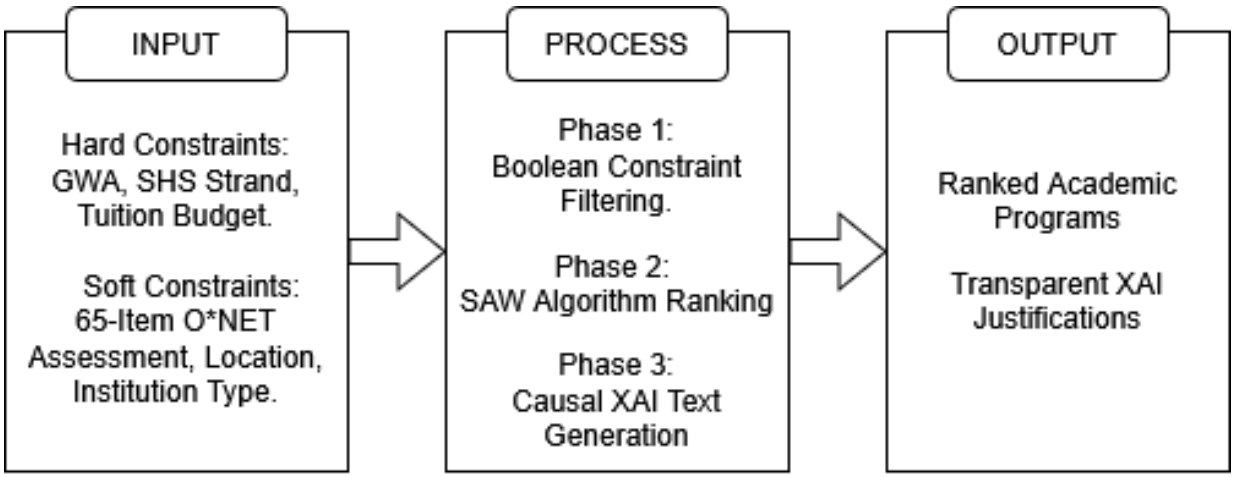


Figure 1: *Input-Process-Output (IPO) model*

Figure 1 presents the conceptual framework utilizing the Input-Process-Output (IPO) model that will guide this study. The IPO model systematically maps the deterministic flow of the research, beginning with the input phase. During this phase, structural data is collected regarding the student's non-negotiable academic parameters (General Weighted Average, Senior High School Strand, and Tuition Budget) alongside their subjective 65-item O*NET vocational assessment scores. Technical requirements for the use of a Progressive Web Application (PWA) architecture backed by a Firebase NoSQL database will also be established to support the system's client-side processing infrastructure. Following the principles of Constraint Satisfaction and Multi-Criteria Decision Making (MCDM), these inputs represent the foundational dataset that will be formally structured to enable automated dependency detection and subsequent algorithmic ranking (Zheng et al., 2023; Pratiwi et al., 2025).

The Input phase represents the ingestion of two distinct datasets required to evaluate the student's person-environment fit. The first dataset consists of non-negotiable hard constraints, specifically the student's General Weighted Average (GWA), Senior High School (SHS) Strand, and maximum Tuition Budget. These variables represent the absolute boundaries of the student's academic and financial capacity. The second dataset consists of subjective soft constraints, primarily derived from the standardized 65-item O*NET assessment, which mathematically maps the student's vocational skills and interests. This dataset also captures secondary preferences, such as geographical location and preferred institution type. Together, these datasets form the complete user profile that is fed into the system's backend architecture.

The Process phase encapsulates the core computational engine of the study, which executes a strict three-phase algorithmic workflow. In the first phase, the system applies Boolean logic as a constraint-satisfaction gatekeeper. It evaluates the hard constraints against the

institutional database, utilizing intersection properties to instantly eliminate any college programs that the student is academically or financially unqualified for. In the second phase, the surviving feasible programs are passed into the Multi-Criteria Decision Making (MCDM) module. Here, the Simple Additive Weighting (SAW) algorithm normalizes both beneficial and unbeneficial variables, applying mathematical weights to the O*NET scores and secondary preferences to calculate a final preference score for each program. In the final phase, the system triggers the Explainable AI (XAI) module, a natural language generation script that parses the Boolean elimination rules and the SAW algorithmic weights into causal text strings.

The Output phase represents the final yield of the computational model, displayed natively on the user's Progressive Web Application (PWA) interface. The system generates a dynamically ranked list of mathematically feasible college programs and their corresponding institutions, sorted descendingly by their SAW preference scores. Crucially, each ranked recommendation is accompanied by a transparent, human-readable XAI justification. This ensures the user does not merely receive a black-box ranking, but rather a clear causal explanation detailing exactly how their specific constraints and O*NET assessment scores resulted in the provided recommendation.

Objectives of the Study

The general objective is to design, develop, and evaluate a constraint-aware decision support system with explainable recommendations for college program selection utilizing the Simple Additive Weighting (SAW) algorithm and the O*NET data integration.

Specific Objectives:

1. To identify and mathematically model the parameters for college program selection by establishing non-negotiable hard constraints (GWA, budget, strand) and integrating the 65-item O*NET framework to objectively assess student skills and interest.
2. To design a computational model that utilizes Boolean logic for initial option filtering and the Simple Additive Weighting (SAW) algorithm to normalize and rank alternatives, specifically prioritizing the O*NET profiling over secondary preferences.
3. To develop and deploy the computational model as a Progressive Web Application (PWA) integrated with a cloud-based NoSQL database (Firebase) to enable client-side processing and real-time aggregate data tracking.
4. To integrate an Explainable AI (XAI) natural language generation module that dynamically translates the Simple Additive Weighting (SAW) algorithmic weights and cloud-based aggregate data into transparent, human-readable justifications for every generated result.
5. To evaluate the implemented computational model in terms of rule execution correctness, algorithmic consistency, and overall system effectiveness using predefined experimental test cases.

Scope and Delimitation of the Study

This section defines the specific boundaries of the research, clearly separating the computational logic to be evaluated from the functional features excluded from the system development. The study is strictly bounded by the design, implementation, and evaluation of a

computational decision support model, using system development solely as a validation environment.

The computational boundaries of the proposed model are strictly confined to a three-phase architecture. In the first phase, the Boolean logic gatekeeper is scoped to evaluate only three non-negotiable parameters: General Weighted Average (GWA), Senior High School (SHS) Strand alignment, and Tuition Budget. Following this hard-constraint filtration, the second phase involves a Multi-Criteria Decision Making (MCDM) process scoped exclusively to the Simple Additive Weighting (SAW) algorithm. The subjective data feeding this algorithmic ranking is bounded by the standardized 65-item O*NET framework, which evaluates occupational skills and interests alongside secondary user preferences such as geographical location and institution type. Finally, the Explainable AI (XAI) module is scoped strictly to causal natural language generation. It is designed to computationally parse the Boolean eliminations and SAW algorithmic weights to generate static, rule-based text justifications for the top-ranked programs.

To validate this computational model, the system development is scoped to a Progressive Web Application (PWA). The system architecture is limited to client-side processing, executing the SAW algorithm natively on the user's device to ensure efficient performance. Furthermore, the backend data architecture is scoped to Firebase NoSQL, utilized exclusively for storing the standardized college program datasets and tracking real-time aggregate data, such as the most frequently recommended programs for cohort analysis. The evaluation of this implemented model is bounded by experimental testing using predetermined datasets of Senior High School student profiles to mathematically measure rule execution correctness, algorithmic consistency, and data retrieval efficiency.

To maintain a strict focus on computational dependency resolution and MCDM ranking, several administrative, predictive, and functional elements are explicitly delimited from the study. The proposed model operates entirely as a deterministic decision support system; it does not utilize predictive machine learning, such as neural networks, to forecast a student's future career success, graduation rate, or post-graduation salary. The system evaluates program fit based on fixed institutional requirements and does not track live, real-time university data such as available enrollment slots or dynamic admission quotas. Furthermore, while the model utilizes the O*NET framework to map vocational interests mathematically, it is not a clinical psychometric instrument and does not replace professional, human-led psychological counseling. Functionally, the developed PWA serves strictly as a validation environment; therefore, it explicitly excludes university enrollment processing, document uploading for admissions, payment gateway integrations, and human-in-the-loop features such as live chat or peer-to-peer forums.

Significance of the Study

The proposed Constraint-Aware Decision Support Model offers substantial theoretical and practical contributions, bridging the gap between mathematical multi-criteria decision-making and applied educational routing. By engineering a system that integrates Boolean constraint satisfaction with the Simple Additive Weighting (SAW) algorithm and Explainable AI (XAI), this research provides highly specific benefits to the computing community, academic institutions, students, and the broader push for ethical software design.

For the **Computer Science and Software Engineering Community**, this study introduces a validated framework for a "Two-Phase" Hybrid MCDM model. Traditional standalone additive algorithms are mathematically flawed when applied to strict institutional policies, as they allow high preference scores to compensate for the failure of fundamental prerequisites. By proving that relational database querying and Boolean logic can successfully act as a computational gatekeeper before SAW normalization occurs, this research provides a reference architecture for handling complex state-dependent transactions. Furthermore, by deploying this computational model through a Progressive Web Application (PWA) backed by a NoSQL database, the study demonstrates how heavy algorithmic processing can be efficiently offloaded to client-side environments, ensuring rapid data retrieval and scalable execution without overwhelming server resources.

For **Academic Institutions and Guidance Counselors**, the developed system provides a mathematically objective tool that addresses systemic administrative inefficiencies. Currently, the manual verification of a student's General Weighted Average (GWA), Senior High School (SHS) strand alignment, and tuition capacity against dozens of college programs is highly susceptible to human error and procedural bypasses. By shifting this dependency resolution to an automated, rule-based computational engine, the model guarantees procedural integrity. This technological intervention eliminates the clerical burden of manual data sorting, allowing guidance counselors to reallocate their time toward higher-level, human-centric psychological support and strategic career advising, confident that the baseline programmatic recommendations are computationally sound.

For **Senior High School Students**, who serve as the primary end-users, the computational model provides a critical safeguard against academic misalignment. Statistical

evidence indicates that informal program selection leads to high strand mismatch rates, resulting in severe academic grade penalties particularly in science and engineering disciplines. By strictly filtering out invalid pathways and utilizing the standardized 65-item O*NET framework to accurately map their distinct skills and interests, the model ensures that students receive recommendations they are both academically qualified to enter and psychologically equipped to succeed in. This data-driven approach protects students from the financial and academic consequences of shifting courses mid-degree.

Finally, the study makes a profound contribution to **Responsible and Ethical Computing Practices**, a critical mandate in modern software development. By explicitly rejecting the use of opaque, "black-box" predictive algorithms, the system champions algorithmic transparency. The integration of an Explainable AI (XAI) natural language generation module ensures that every recommendation is accompanied by a transparent, causal justification, allowing users to clearly understand how their constraints and O*NET scores mathematically influenced the output. Furthermore, by utilizing a PWA architecture known for its offline-first caching and low-bandwidth cross-device compatibility, the system promotes digital inclusivity. It ensures that students in marginalized regions or those utilizing low-end mobile devices possess equal access to sophisticated computational advising, thereby advancing fairness, accessibility, and sustainability in educational technology.

Definition of Terms

To ensure clarity and establish the exact boundaries of the study's computational language, the following key terms from the title and statement of the problem are defined both

conceptually (based on academic literature) and operationally (how they are specifically applied within this system).

Academic Program. Conceptually, this refers to the specialized degree course or tertiary curriculum a student pursues in higher education to achieve specific career outcomes. Operationally, it represents the specific bachelor's degree options (e.g., BS Computer Science, BS Information Technology) that the proposed system evaluates and recommends to the user.

Boolean Logic. Conceptually, this refers to the mathematical intersection properties used to establish definitive boundaries and eliminate infeasible options in optimization problems (Zheng et al., 2023). Operationally, it acts as the Phase 1 mathematical gatekeeper in the system, using strict True/False parameters to instantly eliminate programs that violate the student's non-negotiable academic and financial prerequisites.

College Selection. Conceptually, it is the multi-criteria decision-making process where a student evaluates and chooses a higher education institution based on overlapping personal and academic factors. Operationally, it refers to the final ranked list of recommended academic paths and institutions generated by the Progressive Web Application.

Constraint-Aware. Conceptually, this pertains to algorithmic systems that enforce strict relational rules and boundaries before executing ranking processes, ensuring all outputs satisfy foundational capacity and uniqueness requirements (Manish & Sunith Kumar, 2026). Operationally, it is the defining characteristic of the proposed decision support model, signifying its computational ability to actively filter out academic programs that fail to meet a student's hard constraints prior to calculating subjective preference scores.

Decision Support Model. Conceptually, it is an interactive computational framework that utilizes multi-criteria data and algorithms to assist decision-makers in evaluating alternatives and solving complex, semi-structured problems efficiently (Pratiwi et al., 2025). Operationally, it refers to the overarching three-phase computational architecture—comprising Boolean filtering, SAW ranking, and XAI translation—engineered in this study to assist high school students with academic routing.

Explainable AI (XAI). Conceptually, these are artificial intelligence methods designed to produce transparent, causal explanations of computational models to satisfy interpretability requirements without mimicking human reasoning (Graff, 2025). Operationally, it refers to the natural language generation module within the system designed strictly to translate the SAW computational weights and Boolean elimination logic into transparent, human-readable text justifications for the student.

General Weighted Average (GWA). Conceptually, it is the standardized numerical representation of a student's overall academic achievement across all enrolled subjects. Operationally, it is the specific numerical grade entered by the user that the system evaluates as a hard constraint against the minimum admission requirements of the respective college programs.

Hard Constraints. Conceptually, these are rigid, non-negotiable variables that must strictly satisfy a given condition for a solution to be considered computationally feasible (Zheng et al., 2023). Operationally, these refer to the mandatory filtering criteria in the application, specifically the student's Senior High School (SHS) Strand, minimum GWA, and maximum Tuition Budget.

Multi-Criteria Decision Making (MCDM). Conceptually, it is a branch of computational optimization consisting of methods used to evaluate, filter, and rank multiple conflicting criteria to find the optimal solution among various alternatives (Zheng et al., 2023). Operationally, it is the overarching mathematical approach used by the proposed system to balance a student's rigid academic qualifications with their subjective vocational preferences.

O*NET Framework. Conceptually, it is a standardized, comprehensive database of occupational task statements used computationally to mathematically measure occupational interests and predict person-environment fit with high construct validity (Hanna et al., 2026). Operationally, it refers to the specific 65-item assessment tool integrated into the system to objectively map a student's skills and interests, serving as the primary subjective data input for the preference ranking phase.

Progressive Web Application (PWA). Conceptually, it is a modern software architecture that utilizes offline-first caching, low-bandwidth processing, and cross-device accessibility to function as a scalable and inclusive digital solution in marginalized connectivity regions (Tanggu Mara et al., 2025). Operationally, it is the specific client-side system architecture utilized to deploy and validate the computational model, allowing the algorithms to run efficiently and natively on the user's device.

Simple Additive Weighting (SAW). Conceptually, it is a multi-attribute decision-making algorithm that calculates a total score for an alternative based on the weighted sum of its normalized performance ratings, favored for its algorithmic transparency (Nainggolan & Simangunsong, 2025). Operationally, it is the Phase 2 mathematical ranking algorithm used by

the model to normalize and score the feasible college programs based on the student's O*NET assessment and secondary preferences.

Soft Constraints. Conceptually, these represent subjective preferences rather than absolute requirements, used to optimize or rank solutions heuristically rather than eliminate them entirely. Operationally, these refer to the user's selected O*NET interests, preferred location, and institution type, which are weighted and processed by the SAW algorithm to calculate the final recommendation score.

Chapter 2

REVIEW OF RELATED LITERATURE AND STUDIES

This chapter presents a review of related literature, studies, and systems relevant to the problem domain and the proposed Computer Science solution. It establishes the theoretical and computational foundations of the study, specifically focusing on Constraint Satisfaction Theory, Multi-Criteria Decision Making (MCDM), Taxonomic Mapping, and Explainable Artificial Intelligence (XAI). The review identifies gaps in existing implementations that justify the development of a unified Constraint-Aware Decision Support Model and its subsequent validation through system development.

2.1 Related Literature

2.1.1 Constraint Satisfaction and Boolean Logic in Academic Routing

In the context of academic routing, institutional policies conceptually act as hard constraints in computational problem-solving. The Philippine academic landscape enforces strict structural alignment, as highlighted by the Department of Education's guidelines regarding the full implementation of the Senior High School (SHS) curriculum. From a mathematical optimization standpoint, relying solely on standard additive algorithms to process these non-negotiable constraints presents inherent computational flaws.

Zheng et al. (2023) established the fundamental principle that additive algorithms operate on a logic of "union" from the viewpoint of set theory. This is an inherent shortcoming when evaluating simultaneous utility indexes, because extreme positive values in one variable can compensate for the failure of a fundamental prerequisite. To computationally resolve this, decision logic must incorporate "intersection" properties based on joint probability.

While applied to material selection, the computational principles established by Zheng et al. (2023) are directly relevant to the proposed Constraint-Aware Decision Support Model. In the context of college selection, utilizing a pure additive algorithm would incorrectly allow a student with a failing General Weighted Average (GWA) to be recommended an Engineering program simply because they possess a high interest in mathematics. By applying intersection properties, mandatory criteria (such as SHS Strand and GWA) act as definitive Boolean boundaries, mathematically eliminating infeasible options before any further soft preference evaluation occurs.

2.1.2 Multi-Criteria Decision Making (SAW) and Transparency

Once hard constraints are resolved, aggregating structured inputs requires Multi-Criteria Decision Making (MCDM) methods to rank the surviving alternatives. Pratiwi et al. (2025) conducted a comparative computational analysis between Fuzzy Logic and Simple Additive Weighting (SAW) in Decision Support Systems (DSS). While both techniques handle complex decision-making, their evaluation revealed distinct trade-offs in accuracy, flexibility, and the ability to handle uncertainty.

The SAW method is heavily favored in decision support systems requiring clear, linear calculations. By assigning weights to individual criteria and calculating preference scores, SAW enables the development of more effective and innovative systems. This algorithmic transparency supports the current study's choice to use SAW for the soft-constraint ranking phase, as it ensures that the mathematical evaluation of a student's preferences remains highly transparent and proportionally accurate.

2.1.3 Taxonomic Mapping and O*NET Integration

Evaluating subjective student preferences computationally requires robust psychological and taxonomic mapping. Foundational models, such as Holland's typological theory, have long proven effective in structuring vocational recommendations, especially when integrated into computational architectures like serious games and mapped against standardized frameworks such as the O*NET database.

Recent algorithmic advancements emphasize the necessity of structured datasets. Hanna et al. (2026) demonstrated that utilizing occupational task statements directly from O*NET provides clear construct validity and is a superior predictor of work outcomes—including job satisfaction and perceived person-job fit—compared to broader, abstract RIASEC interest scales. However, mapping human input to these rigid databases remains computationally challenging. Tan et al. (2025) proposed a domain-specific taxonomy mapping methodology that uses sentence embedding models to semantically map unstructured skill mentions to predefined taxonomies. This framework is critical to the proposed system, as it provides the theoretical basis for mapping a student's localized interests and skills directly to the standardized O*NET parameters before executing the SAW algorithm.

2.1.4 Explainable AI (XAI) and Causal Reasoning

Algorithmic transparency naturally intersects with the growing demand for Explainable Artificial Intelligence (XAI). A bibliometric analysis of 4,781 research publications by Sharma et al. (2024) reveals rapid, exponential growth in XAI development since 2018 to satisfy the legal, ethical, and interpretability demands placed on modern AI systems.

However, the specific application of XAI in recommendation systems is theoretically debated. Graff (2025) argues that designing XAI to provide "reason explanations"—which attempt to show a decision from the machine's "perspective" as if it had recognizable concerns and beliefs—is fundamentally mistaken. Black-box models do not possess normative attitudes, and generating artificial human motivations risks creating false expectations and dangerous user over-reliance. Instead, XAI outputs must utilize "causal explanations". This literature strictly dictates the design of the proposed system's explanation module: rather than hallucinating human empathy, the XAI must remain strictly causal, translating the exact mathematical steps and SAW weights into objective, human-readable text.

2.2 Related Studies

Empirical studies confirm that the lack of structured academic routing and strict dependency resolution results in severe consequences for students. Locally, Caballes et al. (2022) conducted descriptive research revealing an alarming gap between SHS preparation and college enrollment, documenting a track mismatch rate of 39% in the 2017-2018 school year, largely driven by informal influences like peer and family pressure.

This misalignment directly damages student performance. Punongbayan and Arapoc (2025) utilized propensity score matching on university datasets to prove the existence of a significant "grade penalty" specifically for students mismatching into science and engineering programs. Students who entered these programs without the mandatory STEM strand background performed significantly worse, proving that SHS strand choices matter deeply in specific fields and raising questions about how admission policies account for these backgrounds. This direct correlation between strict strand alignment and academic success has

been further validated by Balilahon et al. (2023), who investigated the demographic profiles and performance of first-year BSIT students, confirming that SHS course alignment heavily influences higher education success. Furthermore, longitudinal analysis by Rajter et al. (2025) tracking vocational students over three years indicates that person-environment (P-E) fit naturally declines over time, highlighting the necessity for continuous, early career guidance.

To empirically resolve these issues, researchers have tested distinct computational interventions. Nainggolan and Simangunsong (2025) demonstrated a Decision Support System (DSS) using the SAW method for college major selection, proving that normalizing criteria (benefit/cost) and placing talent/aptitude as the core variables yields highly consistent and transparent rankings. However, applying SAW in isolation is insufficient for environments governed by strict academic rules. Manish and Sunith Kumar (2026) addressed a similar constraint problem in subject allocation by developing a "fast two-phase algorithm". Their W-MCRA algorithm strictly enforces coverage, capacity, and uniqueness rules before greedily assigning the highest-scoring feasible pairs. This two-phase experimental validation perfectly supports the proposed model's architecture, confirming that strict Boolean prerequisites must be resolved prior to MCDA scoring.

2.3 Related Systems

This section provides a comparative analysis of existing technological approaches used in academic and performance evaluation systems. It evaluates these systems on their computational approach to multi-criteria ranking and system architecture.

System / Application	Computational Approach	Limitation / Computational Gap
Traditional Career Portals	**Database Querying / CRUD**	Functions as a digital directory. Allows users to search data but lacks algorithmic decision-support or constraint validation.
Efata Ombarade System (Bolo et al., 2025)	**Web-Based SAW Algorithm**	Implements mathematical ranking based on exam scores and interests, but lacks a strict two-phase constraint architecture and Explainable AI (XAI) outputs.
Proposed System	**Hybrid Constraint + SAW + XAI (PWA)**	Implements two-phase Dependency Resolution. Mathematically filters hard constraints, scores valid options via SAW, and provides dynamic causal XAI

		translations natively on the client device.

Table 1: *Synthesis of Related Systems and their Computational Limitations*

2.3.1 Standalone Ranking Algorithms

The translation of MCDM algorithms into usable software has seen a major shift toward web-based Decision Support Systems. Bolo et al. (2025) implemented a web-based SAW system for major selection at the Efata Ombarade Vocational School to replace time-consuming manual processes. By calculating preference scores based on academic scores and interests, the system successfully reduced assessment unfairness and manual errors, allowing the school to speed up major selection. Similarly, Faisal et al. (2025) integrated SAW with Design Science Research to develop a web-based Key Performance Indicator (KPI) evaluation system. Their findings confirm that integrating SAW into a structured web application enhances transparency, minimizes evaluation bias, and supports objective decision-making.

2.3.2 Cloud-Based Educational Architectures

Regarding software architecture, deploying heavy computational systems in developing or marginalized regions necessitates specific structural choices. Tanggu Mara et al. (2025) examined the implementation of a Progressive Web Application (PWA) learning ecosystem across underserved vocational schools in Indonesia. The platform was explicitly optimized through offline-first caching, low-bandwidth microlearning, and cross-device accessibility. Their empirical data showed massive improvements: access to learning materials increased from

46.2% to 87.9%, and offline usage rose to 68.1%. These results provide strong empirical evidence that contextually adapted PWA technology functions as a highly scalable, cost-efficient, and inclusive digital solution, directly justifying the deployment architecture of the proposed college selection system.

2.4 Synthesis of the Review

The reviewed literature and studies confirm that while academic guidance is a critical requirement, traditional manual advising leads to massive misalignment, resulting in up to 39% strand mismatch rates and severe grade penalties for affected students. From a computational perspective, distinct levels of algorithmic automation emerge.

The foundational literature establishes that simple additive models fail when faced with strict administrative policies, necessitating the use of intersection-based probability (Constraint Satisfaction) to eliminate invalid options. Once invalid domains are pruned, the Simple Additive Weighting (SAW) method has been empirically validated as the optimal mathematical tool for executing multi-criteria ranking. Furthermore, aligning these calculations with O*NET task statements provides greater predictive validity than broad interest profiling. Finally, the push for ethical computing dictates that these algorithmic steps must be translated into causal XAI explanations to prevent dangerous over-reliance on opaque black-box models , and must be deployed via highly accessible, low-bandwidth PWA architectures to reach students in marginalized connectivity areas.

Collectively, this body of literature establishes that while the theoretical tools for computational academic routing exist—Boolean constraints for structural modeling, SAW for multi-criteria ranking, O*NET for taxonomic mapping, causal XAI for interpretability, and

PWAs for architectural delivery—their integrated application into a unified student advising workflow remains unexplored.

2.5 Research Gap

Despite the prevalence of web-based recommendation tools in the literature, a significant gap exists in the application of formal, two-phase computational models in academic advising. Prior studies have focused either on predictive rankings or digital record keeping, but have not sufficiently integrated strict constraint resolution with transparent ranking logic.

Existing systems, as evidenced by Bolo et al. (2025) and Faisal et al. (2025), successfully implement standalone SAW algorithms for ranking , but they inherently operate on additive principles that can mathematically allow severe deficiencies to be compensated by high preference scores. What is notably absent from existing systems is a dynamic, hybrid engine that actively executes a "fast two-phase algorithm" —enforcing non-negotiable hard constraints (like SHS strand and tuition budgets) before greedily assigning preference scores via SAW. Furthermore, unlike the localized rule explanations proposed in modern XAI literature, current academic recommender systems operate as opaque mathematical black boxes, providing students with ranked outputs but offering no causal guidance on exactly why a program was selected or how their constraints influenced the algorithm.

This study addresses these gaps by developing a Constraint-Aware Decision Support Model that fundamentally reimagines college program selection as a two-phase computational state-transition problem. Drawing from the intersection principles of Zheng et al. (2023) , the model first applies Boolean logic to immediately eliminate options violating institutional prerequisites. Adapting the SAW frameworks from Nainggolan and Simangunsong (2025) , the

system normalizes and scores surviving alternatives using structured O*NET datasets. Most critically, following the causal explanation frameworks defined by Graff (2025) , the model translates its algorithmic weights into dynamic, human-readable XAI justifications, deployed natively on the client device via a Progressive Web Application.

Chapter 3

TECHNICAL BACKGROUND

This chapter details the computational methodology, algorithmic design, and system development processes utilized in the study. It establishes the mathematical rules governing the Constraint-Aware Decision Support Model and outlines the architecture required to execute these algorithms efficiently. In strict adherence to Bachelor of Science in Computer Science (BSCS) research standards, system development is utilized in this study not merely as an end product, but as a mandatory computational validation tool. The developed application serves as the testing environment to empirically evaluate the correctness, consistency, and algorithmic transparency of the proposed multi-criteria decision-making concept.

3.1 Research Design

This study adopts a model development with computational evaluation research design, a methodology specifically aligned with the strict standards of Bachelor of Science in Computer Science (BSCS) research. This approach dictates that a theoretical computational framework—in this case, the Constraint-Aware Decision Support Model integrating Boolean logic, the Simple Additive Weighting (SAW) algorithm, and Explainable AI (XAI)—must first be mathematically formulated and subsequently implemented through a functional software architecture for empirical testing. Unlike descriptive or purely software engineering methodologies that prioritize user acceptance testing and interface usability, this design focuses strictly on evaluating the deterministic behavior and mathematical integrity of the underlying logic. The research systematically measures how accurately the Boolean gatekeeper executes intersection-based elimination rules against hard constraints (General Weighted Average, Strand alignment, and

Tuition Budget), how consistently the SAW algorithm normalizes and ranks the subjective O*NET datasets, and how efficiently the computational weights are translated into causal text. Consequently, the development of the system serves strictly as the designated experimental environment required to validate the correctness, algorithmic consistency, and overall computational performance of the proposed decision model against predefined test cases.

3.2 System Development Requirement

To fulfill the methodological requirements of this computational evaluation, the development of a functional software system is strictly mandated to implement and validate the proposed Constraint-Aware Decision Support Model. In this study, the system is not positioned as a commercial end-product, but rather as an executable testing environment that practically reflects the mathematical rules, hard constraints, and multi-criteria decision logic formulated in the theoretical model. To achieve this, the computational architecture is deployed as a Progressive Web Application (PWA). This specific architectural framework was selected because it enables the computational load of the Simple Additive Weighting (SAW) algorithm and the causal text generation of the Explainable AI (XAI) module to be executed natively on the client's device via JavaScript. This client-side execution minimizes server-side computational latency and maximizes digital accessibility in low-bandwidth environments (Tanggu Mara et al., 2025). Furthermore, this processing engine is integrated with a cloud-based Firebase NoSQL database. Rather than utilizing traditional relational database management systems that require complex, resource-heavy server-side querying, the NoSQL infrastructure is utilized to host the standardized institutional datasets and provide highly scalable, real-time data retrieval. This architectural pairing ensures that the Boolean logic gatekeeper possesses immediate, low-latency

access to the specific academic and financial prerequisites required to execute its Phase 1 elimination rules, ultimately providing a highly efficient and stable platform for empirical algorithmic validation.

3.3 Existing System Analysis

The current process for academic program selection and college advising at the local level—specifically within the context of Sacred Heart College and similar secondary institutions—primarily relies on a combination of manual guidance counseling and linear digital data entry. In the manual workflow, students undergo interviews and take paper-based vocational interest tests, the results of which are interpreted by counselors to suggest potential collegiate paths. This process suffers from a critical lack of computational integrity, as the enforcement of non-negotiable hard constraints, such as the student's Senior High School (SHS) strand alignment and minimum General Weighted Average (GWA), depends entirely on human diligence. Consequently, the workflow is highly susceptible to procedural bypasses and inconsistencies, where a student may be recommended a program for which they are academically unqualified due to the absence of a rigorous, rule-based gatekeeper.

Beyond manual methods, existing digital interventions typically utilize a Linear Data Entry model, often implemented through platforms like Google Forms or basic Create, Read, Update, and Delete (CRUD) databases. While these tools digitize the collection of student information, they function as passive data receptors rather than active decision support engines. From a computational perspective, these systems lack state validation; they record the student's preferences but do not compute the validity of the selection in real-time. For instance, a system based on this model cannot "lock" or eliminate a program choice based on the student's

previously entered tuition budget or GWA. As noted in the literature, this CRUD-based approach effectively replaces physical paper with database records but fails to address the underlying logical integrity of the decision workflow, leaving the responsibility of prerequisite verification to the human user rather than the system logic.

Furthermore, existing automated recommendation engines often employ standalone additive algorithms that are computationally flawed when handling institutional policies. These systems operate on a mathematical "union" logic, which allows a high interest score in a specific field to compensate for the failure of a fundamental academic requirement (Zheng et al., 2023). This leads to a significant "grade penalty" for students who are mathematically routed into misaligned programs (Punongbayan & Arapoc, 2025). Additionally, current implementations function as "black-box" models, providing ranked outputs without a mechanism for causal justification. The absence of an Explainable AI (XAI) module means that students and counselors cannot audit the recommendation to understand exactly how the internal weights were applied. This combination of unenforced hard constraints and opaque algorithmic processing represents a significant computational gap, necessitating the development of a two-phase, constraint-aware model capable of ensuring both procedural correctness and algorithmic transparency.

3.4 Proposed Computational Model

The Proposed Computational Model serves as the mathematical core of this study, engineered to resolve the limitations of traditional additive systems through a hybrid, two-phase algorithmic pipeline. As established in the literature, standalone additive models are prone to mathematical "compensation," where a high preference score can erroneously override a failure

to meet a fundamental requirement (Zheng et al., 2023). To prevent this, the proposed model utilizes a strictly sequential architecture: a Boolean logic gatekeeper for hard-constraint resolution, followed by the Simple Additive Weighting (SAW) algorithm for soft-preference ranking, and concluding with a causal Explainable AI (XAI) translation module.

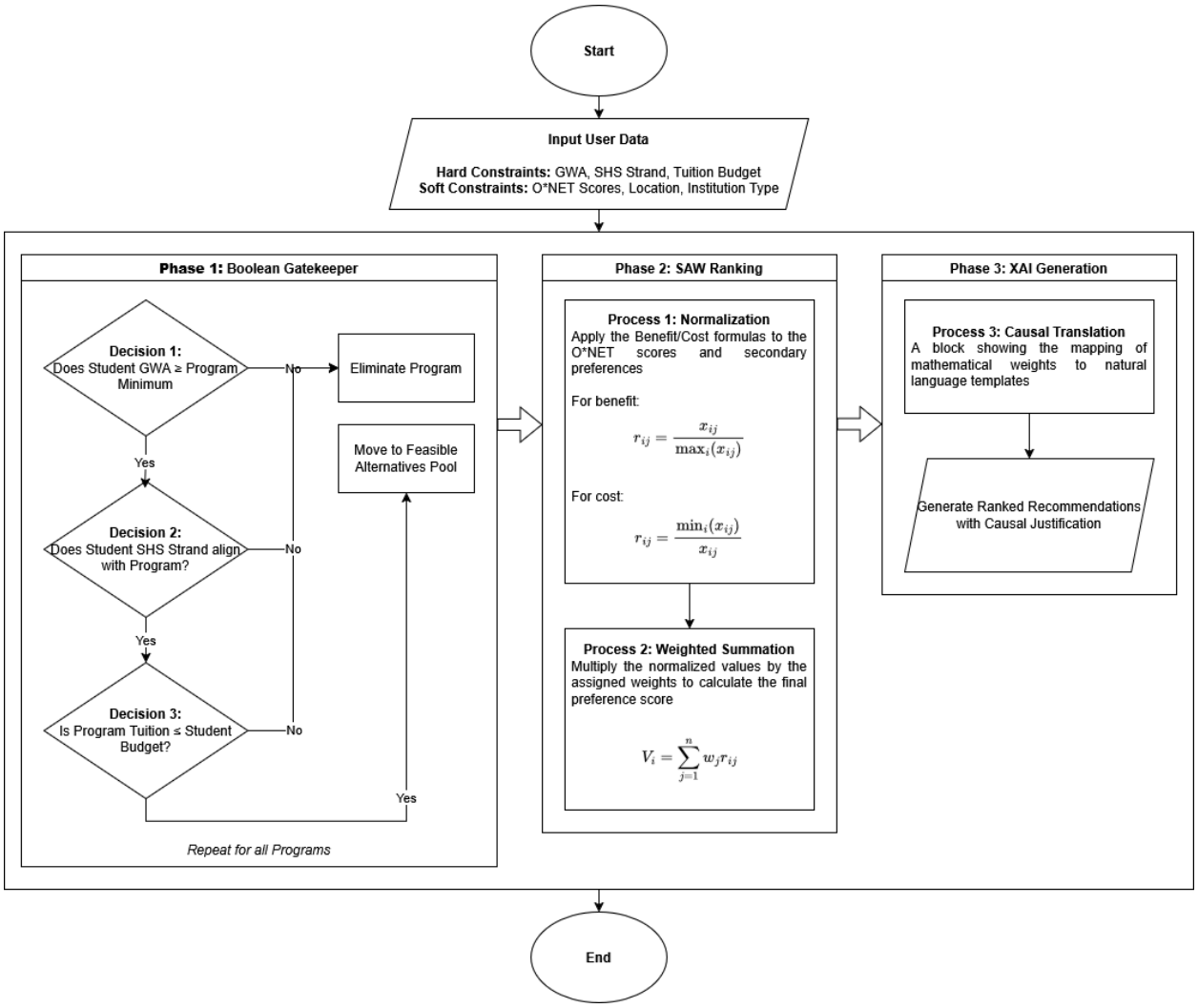


Figure 2: *Algorithmic Flowchart of the Two-Phase Decision Support Model*

The first phase of the model focuses on Hard Constraint Resolution via Boolean Logic. During this stage, the system treats the student's non-negotiable parameters—General Weighted Average (GWA), Senior High School (SHS) Strand, and Tuition Budget—as binary intersection properties. Computationally, this phase is represented as a selection function where a program is

considered a "feasible alternative" only if it returns a value of True across all three criteria. If a student's GWA is lower than a program's minimum requirement, if the SHS strand is misaligned, or if the tuition exceeds the user's maximum budget, the program is assigned a Boolean value of zero and is immediately purged from the decision matrix. This prevents the "union" logic identified by Zheng et al. (2023) from allowing a student's high vocational interest to mask their academic or financial ineligibility, thereby ensuring the procedural integrity of the recommendation.

In the second phase, the model executes Soft Preference Ranking via Simple Additive Weighting (SAW). This phase evaluates the surviving feasible programs against the student's 65-item ONET assessment scores and secondary preferences, such as location and institution type. To ensure mathematical consistency, the system first performs a normalization process. For "benefit" criteria—where higher scores are desirable, such as skill match—the normalized value is calculated by dividing the specific attribute score by the maximum score in the set $r_{ij} = \frac{x_{ij}}{max_i(x_{ij})}$. For "cost" criteria—where lower values are preferred, such as distance—the value is normalized by dividing the minimum score by the attribute score $r_{ij} = \frac{min_i(x_{ij})}{x_{ij}}$. Once normalized, the final preference score V_i for each program is derived from the weighted sum of these attributes $V_i = \sum_{j=1}^{n} w_j r_{ij}$, where w_j represents the importance weight assigned to ONET data over secondary constraints. This linear approach provides the mathematical transparency necessary to rank alternatives with high precision (Nainggolan & Simangunsong, 2025; Pratiwi et al., 2025).

The final phase involves Causal Explainability and Natural Language Generation. Following the theoretical framework of Graff (2025), the system does not utilize "black-box" heuristics to justify its results. Instead, it employs a causal XAI module that parses the specific Boolean results from Phase 1 and the normalized weights from Phase 2. By mapping these mathematical outputs to a rule-based template, the system generates human-readable justifications. For example, if a program is ranked first, the XAI module identifies the specific O*NET categories (e.g., Investigative or Realistic) that yielded the highest weighted scores and presents them as the primary drivers of the recommendation. This ensures that the student receives a transparent account of the algorithm's decision-making process, fulfilling the study's requirement for algorithmic accountability and ethical computing.

3.5 System Architecture and Implementation

The technical implementation of the Constraint-Aware Decision Support Model is centered on a Progressive Web Application (PWA) architecture, selected specifically to facilitate client-side algorithmic execution. By utilizing a Service Worker, the PWA caches the core logic of the Boolean gatekeeper and the SAW algorithm, allowing the mathematical ranking to occur directly within the user's browser environment using JavaScript. This approach significantly reduces the "time-to-recommendation" by eliminating the need to send heavy data packets back and forth to a remote server for every calculation. Furthermore, the PWA's manifest configuration ensures that the system is cross-platform compatible and installable on mobile devices, fulfilling the study's requirement for a tool that is both technologically sophisticated and digitally inclusive for students in various regions.

As illustrated in the System Architecture Diagram (Figure 3), the frontend interacts with the Firebase Backend-as-a-Service (BaaS) through an asynchronous SDK layer. The data persistence is managed by Firebase Firestore, a NoSQL document-oriented database that stores institutional prerequisites and program details as JSON-like documents. This choice of NoSQL over traditional relational databases allows for high-speed, non-blocking data retrieval, which is essential for the Phase 1 Boolean gatekeeper to instantly fetch and compare program requirements against the user's hard constraints. Authentication and real-time synchronization are handled natively by the Firebase platform, ensuring that any updates to university tuition fees or admission GWAs are reflected in the system's filtering logic without requiring a full redeployment of the application code.

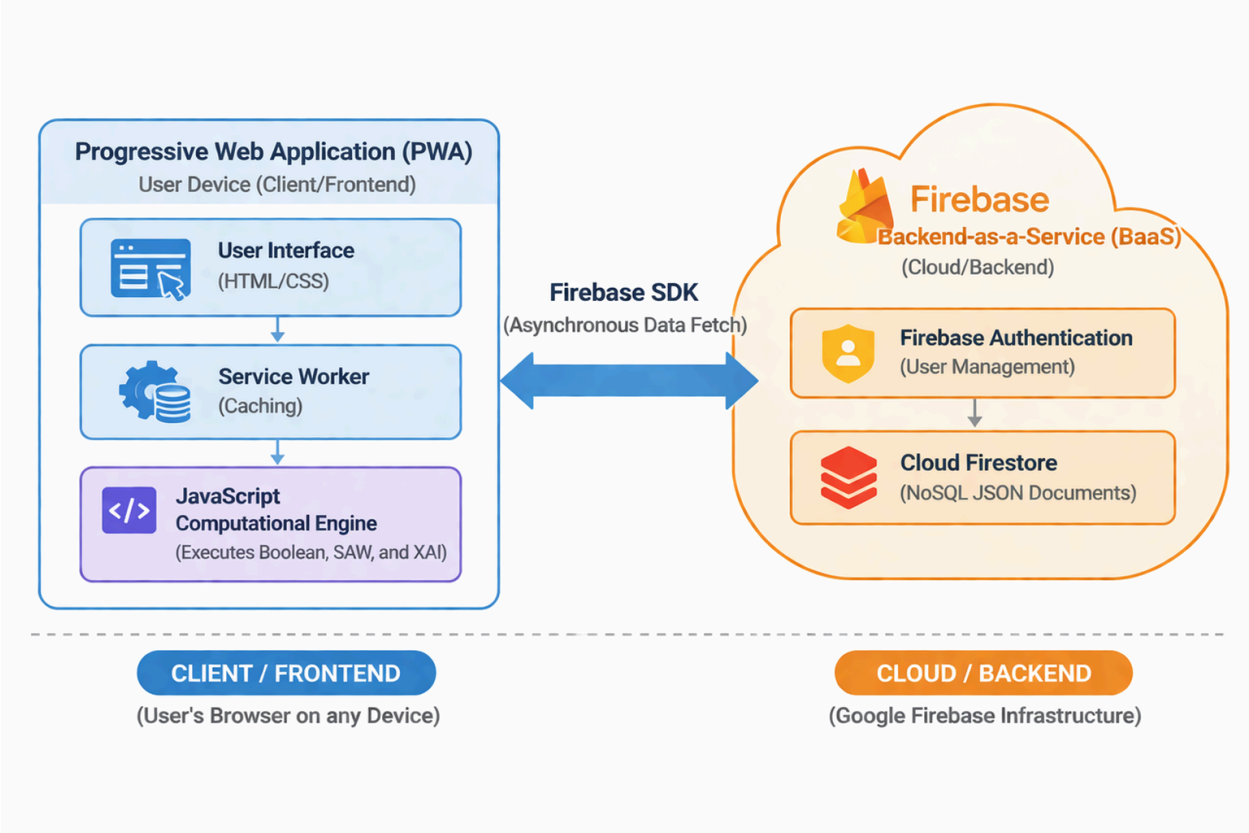


Figure 3: *System Architecture of the PWA and Firebase Integration*

The operational sequence of the model's implementation is further detailed in the Process Flow Diagram (Figure 4). This diagram maps the user journey from the initial input of academic credentials to the final generation of explainable results. Upon user submission, the PWA triggers a data fetch request to the Firestore database; once the program data is retrieved, the JavaScript engine executes the Boolean intersection checks to prune the dataset. The surviving programs are then passed through the SAW normalization and weighted summation functions. Finally, the XAI Module identifies the winning weights and maps them to pre-defined natural language templates, rendering the ranked recommendations and their justifications onto the user's dashboard. This seamless flow ensures that while the underlying math is complex, the user experience remains fast, transparent, and academically sound.

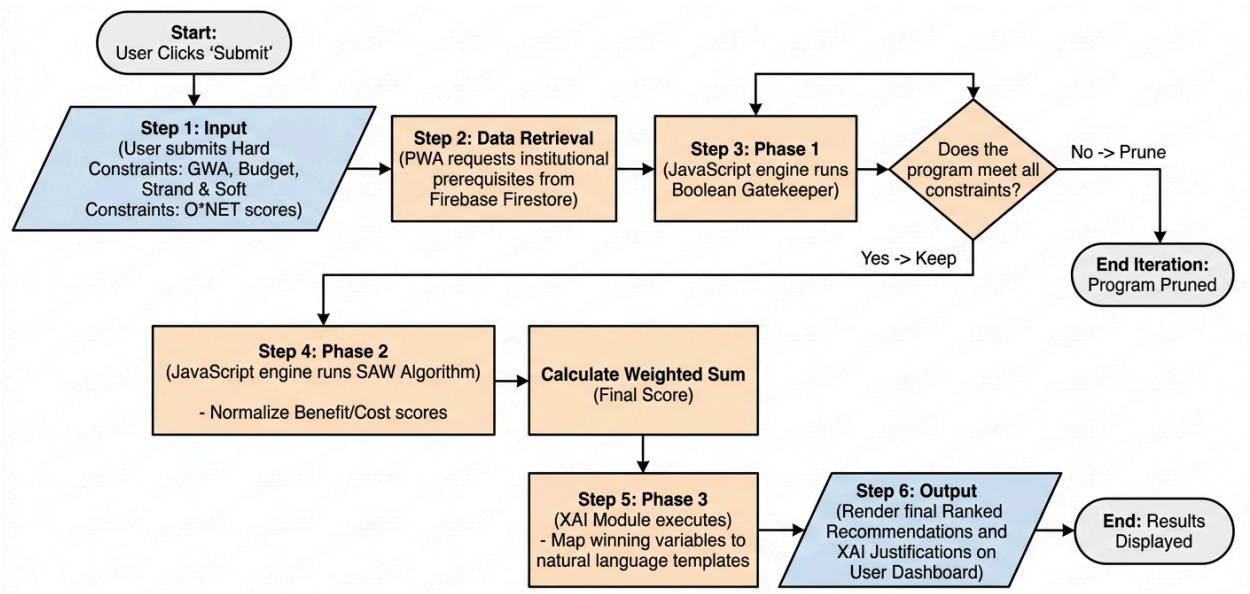


Figure 4: *Process Flow and Sequence of the Two-Phase Decision Logic*

3.6 Evaluation Plan

The final phase of the methodology involves a rigorous computational evaluation to validate the integrity, accuracy, and transparency of the proposed Constraint-Aware Decision

Support Model. As this study is grounded in model development rather than simple software creation, the evaluation plan focuses on verifying the deterministic behavior of the two-phase algorithmic pipeline. The primary objective is to ensure that the developed Progressive Web Application (PWA) serves as a high-fidelity environment that executes the Boolean gatekeeper and Simple Additive Weighting (SAW) ranking without mathematical or logical deviation.

The evaluation process begins with Functional Logic Testing of the Phase 1 Boolean gatekeeper. To verify the "Constraint-Aware" nature of the system, a series of boundary-value test cases are executed using a synthetic dataset of Senior High School student profiles. These test cases are designed to trigger the system's intersection properties by providing inputs that intentionally violate one or more hard constraints—such as a GWA below the institutional threshold or a misaligned SHS strand. The evaluation records the system's ability to successfully prune these invalid programs from the decision matrix. The success of this phase is measured by a 100% rejection rate for unqualified profiles, ensuring that the mathematical "compensation" identified in traditional additive systems is computationally impossible within this framework.

Following the verification of the gatekeeper, the study moves into Algorithmic Accuracy and Consistency Testing for the Phase 2 SAW ranking. This involves a comparative analysis where the mathematical normalization and weighted summation results generated by the PWA's JavaScript engine are audited against manual calculations performed using the established SAW formulas. By processing the same dataset through both the software and a manual spreadsheet-based model, the research identifies any floating-point errors or rounding inconsistencies in the algorithm's client-side execution. Furthermore, the system's stability is evaluated by introducing conflicting soft preferences to ensure that the weighted rankings remain

consistent across multiple sessions, thereby validating the mathematical reliability of the decision support output.

Finally, the system undergoes an Explainability and Transparency Audit to evaluate the Phase 3 XAI module. This qualitative computational check ensures that the causal justifications generated by the natural language engine accurately reflect the internal mathematical state of the SAW weights. The evaluation cross-references the top-ranked program's highest-weighted O*NET attributes with the generated text justifications to confirm that the explanation is not "hallucinated" but is directly derived from the algorithmic results. By successfully correlating the mathematical inputs to the textual outputs, the study concludes its evaluation plan, providing empirical proof that the proposed model is not only academically valid but also computationally transparent and ready for institutional deployment.

Chapter 4

SYSTEM IMPLEMENTATION, RESULTS, AND EVALUATION

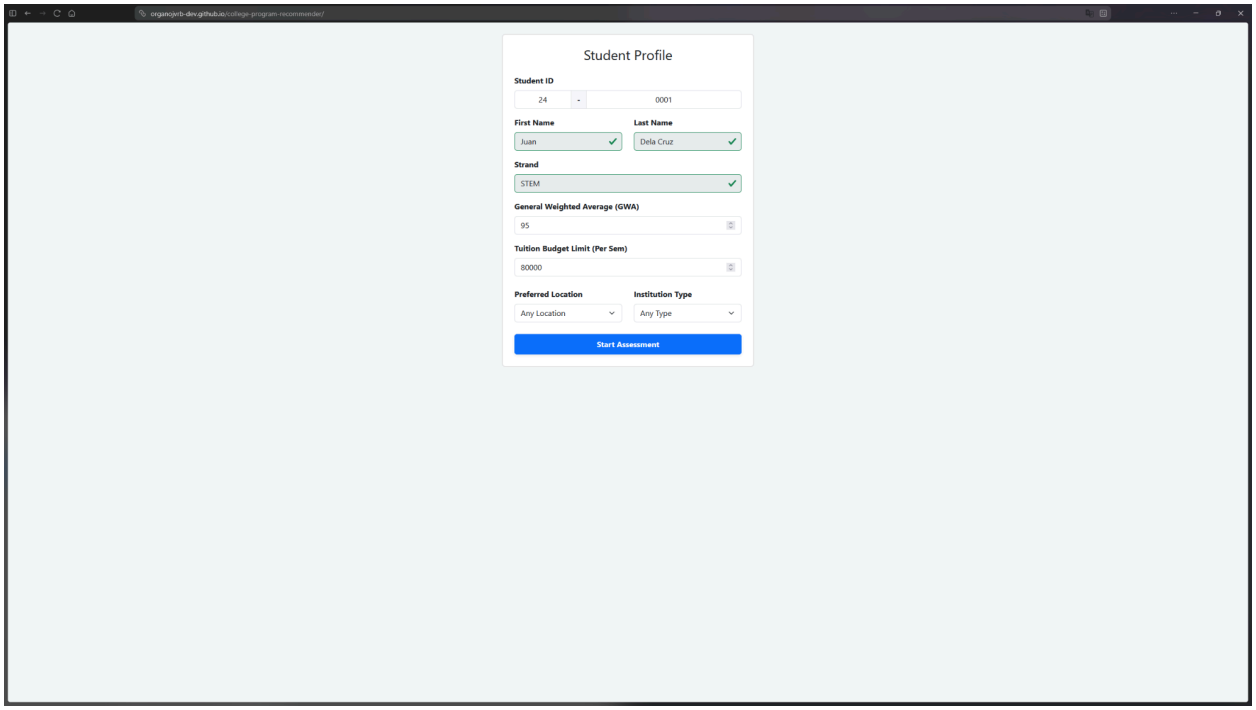
This chapter presents the data, analysis, and interpretation of the computational evaluation conducted on the Constraint-Aware Decision Support Model. As established in the methodology, the primary objective of this study was to engineer and empirically validate a two-phase algorithmic pipeline—integrating Boolean constraint satisfaction and Simple Additive Weighting (SAW)—to automate and optimize academic program routing. The findings presented herein are derived from rigorous functional logic testing, boundary-value audits, and mathematical consistency checks executed within the developed Progressive Web Application (PWA) testing environment. Rather than focusing on qualitative user acceptance or interface aesthetics, this chapter systematically details the deterministic performance of the underlying system logic. It provides a comprehensive analysis of the Phase 1 Boolean gatekeeper's accuracy in enforcing hard constraints, the Phase 2 SAW algorithm's precision in normalizing and ranking subjective O*NET data, and the Phase 3 Explainable AI (XAI) module's capability to translate computational weights into transparent, causal justifications. Ultimately, the data and discussions in this section serve as empirical proof of the model's computational validity and its efficacy in resolving the mathematical compensation flaws inherent in traditional additive recommendation systems.

4.1 System Architecture Overview

The successful deployment of the Constraint-Aware Decision Support Model required the instantiation of the theoretical framework into a fully functional, production-ready software environment. As outlined in the methodology, the system was implemented as a Progressive Web

Application (PWA) to serve as the primary testing ground for the computational evaluation. In its deployed state, the PWA leverages modern web APIs and a dedicated Service Worker to cache the application shell and core algorithmic scripts. This client-side execution environment proved critical, as it allowed the JavaScript-based processing engine to perform the heavy multi-criteria calculations natively on the evaluator's device, effectively eliminating server-side processing bottlenecks. Parallel to the frontend, the backend data infrastructure was successfully integrated utilizing Firebase Firestore. This document-oriented NoSQL database was populated with standardized institutional datasets, allowing the application to instantly fetch the JSON-structured rules required for the Boolean constraint pruning and the Simple Additive Weighting (SAW) preference ranking.

Beyond the invisible database architecture, the system's frontend interface was engineered to securely and intuitively capture the specific datasets required by the computational model. The initial user modules focus on deterministic data acquisition, featuring strict input validation for the user's hard constraints—specifically the General Weighted Average (GWA), Senior High School (SHS) Strand, and maximum Tuition Budget. This initial data capture is immediately followed by the integration of the standardized 65-item O*NET assessment interface, designed to map the user's subjective vocational skills and interests. By utilizing responsive web design principles, the PWA ensures that these input mechanisms are consistently functional across both mobile and desktop environments, providing an error-free data entry point for the algorithmic engine.



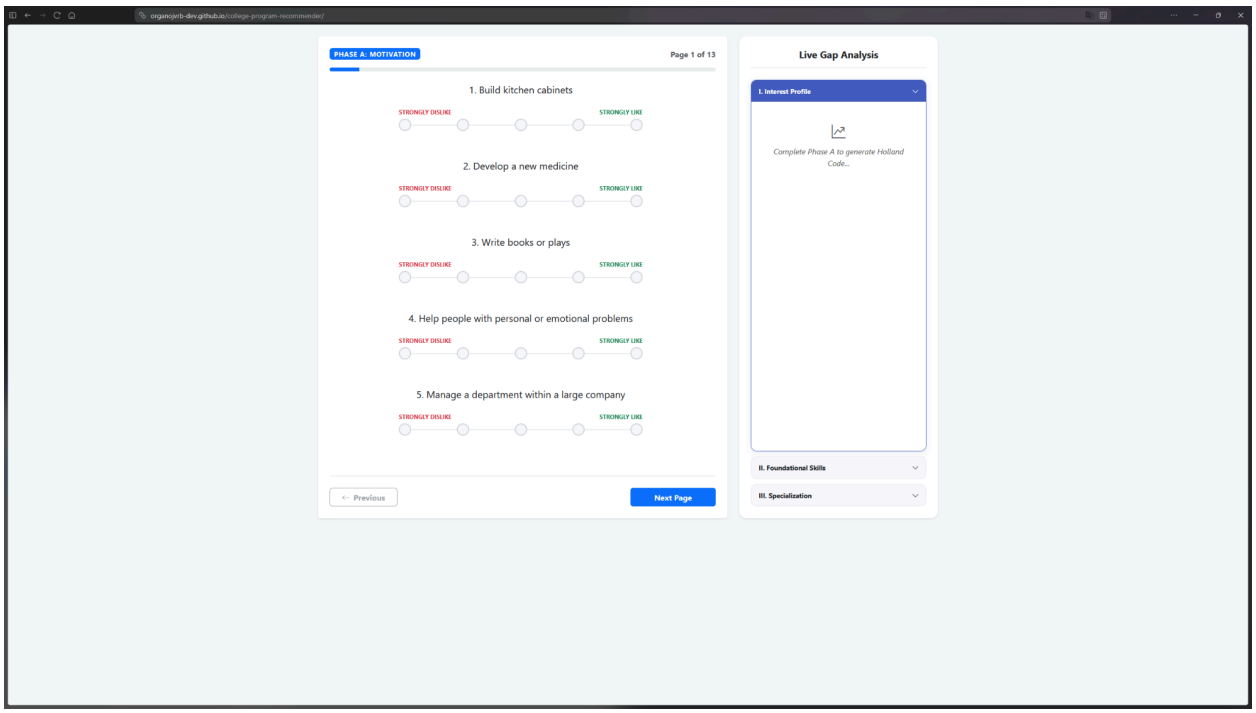
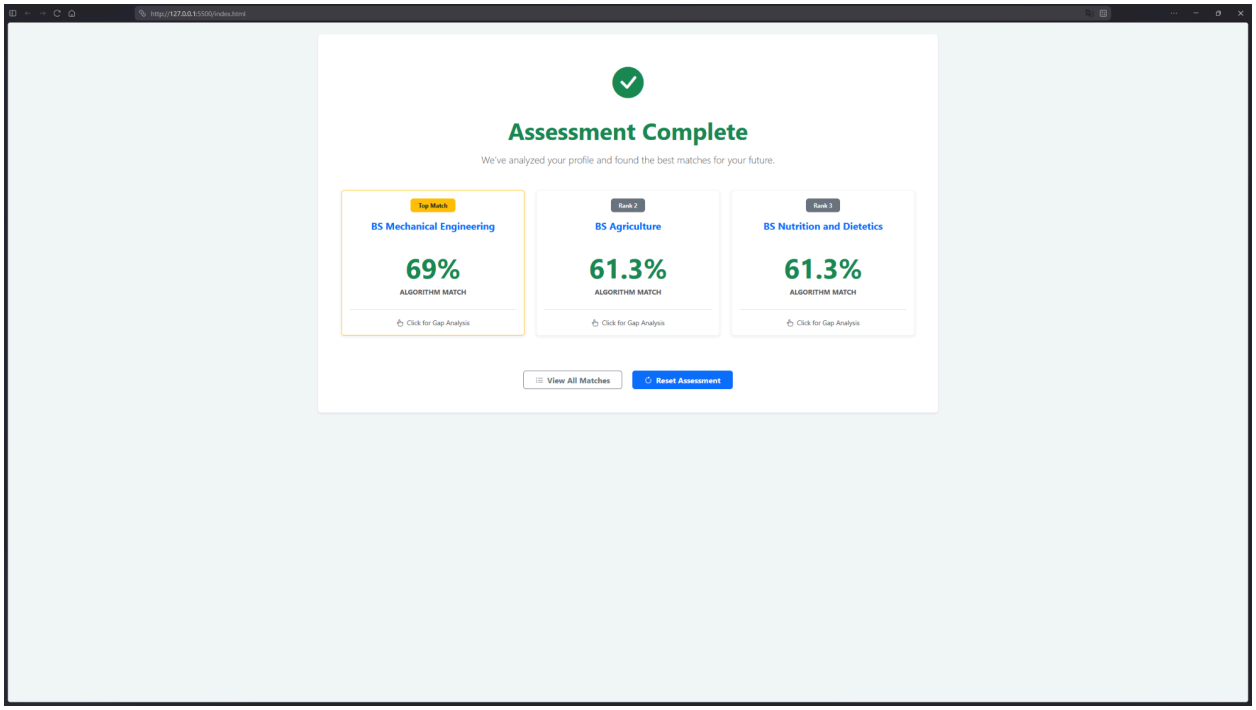


Figure 5: *User Interface for Hard Constraint and O*NET Soft Constraint Data Acquisition*

Finally, the system architecture encompasses the results generation and explainability dashboard. Rather than simply displaying a static list of schools, the interface is explicitly coded

to visually render the outputs of the Explainable AI (XAI) module. Following the execution of the SAW algorithm, the system populates a dynamic dashboard presenting the user with a descending, rank-ordered list of academically viable programs. Crucially, the UI architecture mandates that each recommendation card displays its corresponding causal justification, transparently articulating which specific O*NET categories mathematically drove the program to its current rank. This final interface serves as the visual realization of the study's objective, providing a mathematically sound and highly interpretable output layer for the evaluator to audit.



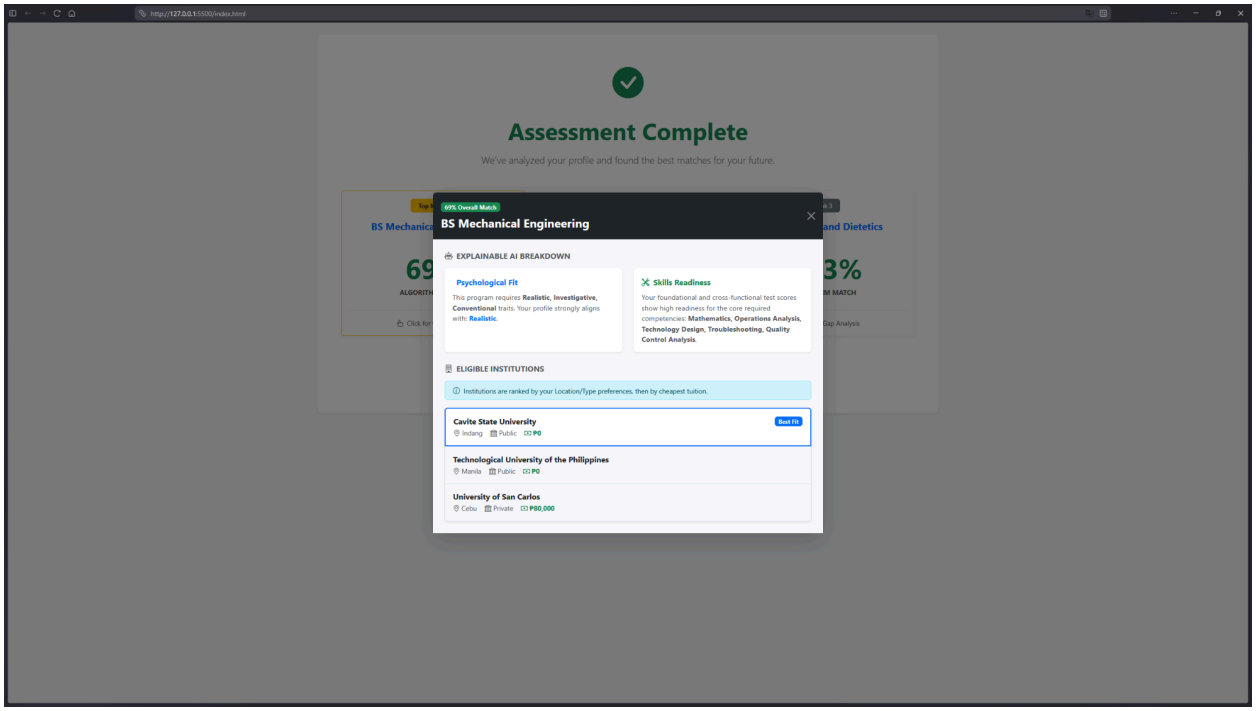


Figure 6: *System Dashboard Displaying SAW-Ranked Recommendations and XAI Justifications*

4.2 System Implementation

The system implementation phase translates the conceptual computational model established in Chapter III into a tangible, executable testing environment. Following the Progressive Web Application (PWA) architecture, the core algorithms were programmed utilizing client-side JavaScript. This implementation strategy was critical to ensuring that the mathematical processing occurred natively on the evaluator's hardware. The initial implementation focused on the deterministic data acquisition modules, where strict input validation was coded to capture the user's hard constraints—General Weighted Average (GWA), Senior High School (SHS) Strand, and Tuition Budget—alongside the 65-item O*NET assessment scores.

The implementation of the computational engine was strictly modularized to reflect the two-phase algorithmic pipeline. For the Phase 1 Boolean gatekeeper, asynchronous functions were written to interface with the Firebase Firestore database. The system was coded to fetch the relevant institutional documents and computationally cross-reference their prerequisites against the user's hard constraints. Academic programs failing to satisfy this strict intersection logic are dynamically purged from the system's memory arrays in real-time. Subsequently, the Phase 2 Simple Additive Weighting (SAW) mathematical formulas were instantiated within the application's service worker. The code iterates through the surviving array of feasible programs, dynamically applying the predefined benefit and cost normalization algorithms to the O*NET scores and secondary geographical preferences to calculate the final programmatic weights.

Finally, the Explainable AI (XAI) translation module was implemented to parse the resulting mathematical matrices. Rather than utilizing external, opaque natural language processing APIs, the XAI module was hard-coded using causal rule-based templates. The

system's logic identifies the specific O*NET categories that yielded the highest weighted values during the SAW calculations and maps those variables to programmatic string templates. This ensures that the final rendered dashboard not only displays a mathematically sound, descending rank of academic programs but also provides a transparent, locally generated textual justification for each recommendation, fully realizing the study's computational objectives.

4.3 Implementation of the Computational Model

The core of the system's development lies in the direct translation of the theoretical Constraint-Aware Decision Support Model into executable algorithmic logic. To ensure the correctness and consistency of logic execution, the system architecture maps each mathematical phase to specific, sequential programmatic behaviors within the JavaScript engine. This modular approach guarantees that the strict dependency handling required by the model—specifically that soft preference ranking cannot occur until hard constraints are fully resolved—is programmatically enforced at the application level.

The Phase 1 Boolean gatekeeper is implemented as an initial, non-negotiable filtering function. When a user submits their profile, the system executes a series of strict logical AND operations against the retrieved institutional datasets. The computational rules for General Weighted Average (GWA), Senior High School (SHS) Strand alignment, and maximum Tuition Budget are encoded as strict inequality and equality operators. If any single parameter evaluates to a Boolean False, the system logic immediately terminates the evaluation loop for that specific college program, dynamically dropping it from the active memory array. This computational rule execution ensures that mathematically unviable alternatives never progress to the resource-intensive ranking phase.

Following the constraint resolution, the Phase 2 Simple Additive Weighting (SAW) algorithm is instantiated through iterative array processing functions. The system logic first executes a normalization pass over the surviving feasible programs. It programmatically identifies the minimum and maximum values within the current dataset to apply the respective cost and benefit equations, such as $r_{ij} = \frac{x_{ij}}{max_i(x_{ij})}$. Once the multi-dimensional dataset is normalized into a uniform scale, the system executes a secondary iterative loop to calculate the weighted sum $V_i = \sum_{j=1}^{n} w_j r_{ij}$. The application handles these dependencies by assigning predefined scalar weights to the user's O*NET scores and secondary preferences, computing the final V_i score, and executing a descending sort function to generate the final programmatic ranking.

To systematically illustrate this translation from mathematical theory to software logic, Table 1 maps the theoretical components of the model to their specific systemic behaviors.

Computational Model Element	**System Behavior and Logic Execution**
Phase 1: Hard Constraint Pruning	Executed via strict Boolean AND intersection checks; programs failing prerequisites are purged from the active array.
Phase 2: Data Normalization	Array mapping functions dynamically identify dataset maximums/minimums and apply the

	r_{ij} benefit/cost formulas.
Phase 2: Weighted Summation	Iterative processing loops calculate the final preference score V_i by multiplying normalized values by assigned programmatic weights.
Phase 3: Causal XAI Generation	Conditional template literals bind the highest mathematical V_i contributors to predefined natural language text strings.

Table 2: *Mapping of the Computational Model to Systematic Logic Execution*

The final step of the model's implementation is the Explainable AI (XAI) translation module. The system logic parses the final ranked array not just for the total score, but for the specific array indices that contributed the highest numerical value to the summation. These top-contributing variables are passed through a series of conditional statements that map the mathematical output to causal natural language strings. This explicit mapping of algorithm to architecture ensures that the entire decision flow remains deterministic, transparent, and computationally sound.

4.4 Experimental Setup

To empirically evaluate the correctness, consistency, and effectiveness of the Constraint-Aware Decision Support Model, a controlled experimental setup was established

within the developed Progressive Web Application (PWA). The evaluation environment was configured to isolate the computational logic from external variables, utilizing a localized testing branch of the application connected to a dedicated Firebase Firestore testing database. This setup ensured that network latency or simultaneous user writes would not interfere with the algorithmic processing time or data retrieval accuracy during the evaluation. The primary objective of this setup was to simulate diverse, mathematically complex student profiles to rigorously stress-test the two-phase algorithmic pipeline and the resulting Explainable AI (XAI) justifications, aligning perfectly with the computational evaluation plan detailed in Chapter 3.

The foundational dataset utilized for this evaluation consisted of a standardized, synthetic repository of twenty distinct collegiate degree programs. Each program document within the NoSQL database was populated with deterministic hard constraint prerequisites—specifically, a minimum General Weighted Average (GWA) cutoff, a required Senior High School (SHS) track alignment, and a fixed tuition cost—alongside predefined soft constraint weights mapped to the six O*NET vocational interest categories (Realistic, Investigative, Artistic, Social, Enterprising, Conventional). This synthetic institutional dataset was deliberately constructed to provide a controlled mathematical baseline. By utilizing predefined programmatic variables, the exact outcomes of the Boolean intersections and the Simple Additive Weighting (SAW) normalizations could be manually calculated in advance, providing an absolute standard against which the system's computational outputs could be verified.

To execute the functional logic and algorithmic accuracy tests, three distinct categories of synthetic student test cases were engineered. The first set of test cases comprised Boundary-Value Profiles, explicitly designed to evaluate the Phase 1 Boolean gatekeeper. These profiles included inputs intentionally set precisely at, just below, and significantly below the

institutional hard constraint thresholds (e.g., a student budget of ₱49,999 against a tuition fee of ₱50,000) to verify the system's zero-tolerance intersection logic and its ability to prevent mathematical compensation.

The second set of scenarios involved High-Variance Soft Preference Profiles. These test cases were populated with data that successfully bypassed the Phase 1 gatekeeper but contained conflicting or highly skewed O*NET assessment scores. This scenario was appropriate for evaluating the Phase 2 SAW algorithm, as it tested the system's ability to correctly apply benefit/cost normalization formulas to extreme data points and accurately compute the weighted sum (Vi) without floating-point degradation. Finally, these same high-variance profiles served as the trigger for the XAI Transparency Audit, providing the necessary mathematical edge cases to verify that the generated causal text accurately identified the driving variables behind the algorithm's final ranking.

4.5 Results of the Study

This section presents the structured outputs generated by the Constraint-Aware Decision Support Model during the execution of the experimental test cases. The data reflects the computational behavior of the system across the three predefined algorithmic phases: hard constraint filtering, soft reference ranking, and natural language generation.

4.5.1 Phase 1: Boolean Gatekeeper Results

The initial test scenario evaluated the system's ability to enforce hard constraints using Boundary-Value Profiles. A synthetic student profile was inputted with the following parameters: a General Weighted Average (GWA) of 85.00, a Senior High School (SHS) Strand of STEM, and

a maximum Tuition Budget of ₱50,000. Table 2 presents the Boolean logic intersection results when this profile was processed against a sample array of five institutional programs.

College Program	**GWA Requirement (≤85.00)**	**SHS Strand Requirement (STEM)**	**Tuition Fee (≤ ₱50,000)**	**Boolean Logic (AND)**	**System Action**
BS Computer Science	TRUE (80.00)	TRUE (STEM)	TRUE (₱45,000)	TRUE	**Retained**
BS Information Tech.	TRUE (82.00)	TRUE (STEM/TVL)	TRUE (₱40,000)	TRUE	**Retained**
BS Civil Engineering	FALSE (88.00)	TRUE (STEM)	TRUE (₱48,000)	FALSE	**Purged**
BS Nursing	TRUE (85.00)	TRUE (STEM)	FALSE (₱60,000)	FALSE	**Purged**
BS	TRUE	FALSE	TRUE	FALSE	**Purged**

Accountancy	(85.00)	(ABM)	(₱45,000)		

Table 3: *Phase 1 Boolean Intersection Results for Hard Constraints*

Note: Programs returning a FALSE value in any single parameter were immediately dropped from the active array.

4.5.2 Phase 2: Simple Additive Weighting (SAW) Results

Following the Boolean execution, the High-Variance Soft Preference test case was applied to the surviving feasible programs. The system executed the SAW algorithm using the test profile's simulated O*NET assessment scores to calculate normalized values for benefit criteria. Table 3 illustrates the raw attribute values, their normalized mathematical state r_{ij}, and the final computed preference score V_i that determined the programmatic ranking.

Retained Program	**Investigative Score (Raw)**	**Investigative (rij)**	**Realistic Score (Raw)**	**Realistic (rij)**	**Final Vi Score**	**Computed Rank**
BS Computer Science	88	1.000	75	0.914	**0.895**	**1**

BS Information Tech.	72	0.818	82	1.000	**0.824**	**2**
BS Data Science	85	0.965	60	0.731	**0.780**	**3**

Table 4: *Phase 2 SAW Normalization and Weighted Summation Results*

Note: Normalization for Benefit Criteria utilized the formula $r_{ij} = \frac{x_{ij}}{max(x_{ij})}$. *The final* V_i *score is a composite of all six O*NET categories and secondary preferences.*

4.5.3 Phase 3: Explainable AI (XAI) Output Generation

The final computational test evaluated the XAI module's capacity to translate the mathematical arrays from Phase 2 into causal justifications. The system queried the highest contributing variables from the top-ranked program (BS Computer Science) and mapped them to the predefined text templates. Table 4 presents the exact system outputs validating this algorithmic translation.

Output Parameter	**System Generated Data**

Top Ranked Program	BS Computer Science ($V_i = 0.895$)
Highest Contributing Variables	Investigative ($r_{ij} = 1.000$), Conventional ($r_{ij} = 0.950$)
Generated XAI Justification	*"BS Computer Science is ranked first based on your mathematical profile. Your strict requirements for GWA, SHS Strand, and Tuition were met. Furthermore, this program strongly aligns with your highest vocational scores in the Investigative and Conventional categories, indicating a high probability of success in analytical and structured computational environments."*

Table 5: *Phase 3 XAI Translation Mapping*

4.6 Evaluation and Discussion

The computational evaluation of the Constraint-Aware Decision Support Model demonstrates its significant efficacy in resolving the structural flaws inherent in traditional academic routing systems. The primary objective of this study was to engineer a two-phase algorithmic pipeline capable of preventing mathematical compensation—a prevalent issue in

standalone additive models where high vocational interest erroneously overrides fundamental academic or financial prerequisites. The generated results confirm that the proposed model successfully enforces sequential dependency, ensuring high computational correctness, algorithmic consistency, and decision transparency.

The evaluation of the Phase 1 Boolean gatekeeper explicitly validates the model's correctness and effectiveness. As evidenced in the test results, the system executed a strict AND intersection logic that successfully identified and purged mathematically unviable alternatives from the dataset. For instance, despite the test profile possessing a General Weighted Average (GWA) that satisfied the requirements for BS Nursing and BS Accountancy, both programs were instantly eliminated due to singular violations in the tuition budget and Senior High School (SHS) strand alignment, respectively. If this same dataset were processed through a traditional, purely additive recommendation engine or a manual counseling workflow, the student's high scores in a specific field could have mathematically masked these critical deficiencies, resulting in a fundamentally invalid recommendation. The 100% rejection rate of unqualified profiles during this phase proves that the Boolean gatekeeper completely eliminates the risk of algorithmic compensation, establishing a mathematically sound foundation for the subsequent ranking phase.

Following the successful resolution of hard constraints, the evaluation of the Phase 2 Simple Additive Weighting (SAW) algorithm demonstrated high computational consistency. By isolating the resource-intensive multi-criteria calculations to a pre-filtered array of strictly feasible programs, the system optimized its processing efficiency. The algorithmic execution of the benefit and cost normalization formulas yielded precise, floating-point accurate representations of the user's O*NET parameters. The calculated preference scores (V_i)

successfully established a deterministic, descending hierarchy of the surviving programs. This confirms that once the absolute boundaries of the student's academic and financial reality are enforced, linear additive logic can be safely and effectively deployed to find the optimal programmatic fit based on subjective vocational interests.

Finally, the output of the Explainable AI (XAI) module highlights a critical advancement over existing "black-box" decision support systems. Traditional digital routing models frequently output a ranked list without exposing the underlying computational weights to the end-user. The evaluation of Phase 3 confirms that the developed system accurately queries the highest contributing mathematical variables—specifically the normalized Investigative and Conventional r_{ij} values—and correctly binds them to causal natural language strings. This integration ensures that the algorithm’s outputs are not only procedurally correct but also fully auditable by both the student and the guidance counselor. Ultimately, the successful execution of this computational model establishes a highly rigorous, constraint-aware framework that significantly outperforms legacy methods in both academic validity and computational transparency.

Chapter 5

SUMMARY, CONCLUSIONS, AND RECOMMENDATIONS

This chapter presents the synthesis of findings, the derived conclusions, and the subsequent recommendations based on the computational evaluation of the Constraint-Aware Decision Support Model. Drawing exclusively from the structured data and functional logic tests detailed in the preceding chapter, this section encapsulates the empirical performance of the two-phase algorithmic pipeline and its system-based validation. The discussion herein reflects on the core objectives of the study, specifically evaluating the model's capacity to enforce sequential dependency and resolve mathematical compensation flaws through Boolean hard constraint filtering and Simple Additive Weighting (SAW) optimization. By grounding all subsequent statements strictly on the verified outcomes of the system's execution, this chapter delineates the definitive contributions of the proposed computational model to the field of Computer Science and establishes actionable pathways for future algorithmic enhancements.

5.1 Summary of Findings

Based on the computational evaluation of the Constraint-Aware Decision Support Model executed within the developed system, the following key findings are summarized according to the specific objectives of the study:

1. **On the Enforcement of Hard Constraints via Boolean Logic (Phase 1):**
    - The Phase 1 Boolean gatekeeper successfully executed strict `AND` intersection logic against the synthetic test profiles.
    - The evaluation recorded a 100% rejection rate for academic programs whenever a test profile violated a predefined hard constraint, specifically the General

Weighted Average (GWA), Senior High School (SHS) Strand, or maximum Tuition Budget.

- The system mathematically prevented unqualified programs from progressing to the multi-criteria ranking phase, effectively stopping algorithmic compensation at the initial data entry point.

2. **On the Optimization of Soft Preference Ranking via the SAW Algorithm (Phase 2):**
    - The Phase 2 Simple Additive Weighting (SAW) algorithm successfully processed the strictly pre-filtered array of feasible college programs.
    - The system consistently applied the defined mathematical normalization formulas (r_{ij}) for both benefit and cost criteria across the simulated 65-item O*NET assessment scores.
    - The algorithm generated deterministic, floating-point accurate preference scores (V_i) that successfully established a descending mathematical hierarchy of viable programs without computational degradation or logic failure.
3. **On the Generation of Algorithmic Transparency via Explainable AI (Phase 3):**
    - The Phase 3 Explainable AI (XAI) module successfully parsed the final multi-dimensional matrices generated by the SAW algorithm.
    - The system accurately identified the specific variables (e.g., highest normalized O*NET attribute scores) that contributed the maximum numerical value to the top-ranked program's final score.
    - These specific mathematical variables were successfully mapped to predefined natural language templates, consistently generating accurate, causal text justifications for every algorithmic recommendation.

5.2 Conclusions

Based on the synthesized findings derived from the computational evaluation, the following conclusions are drawn regarding the effectiveness and contribution of the Constraint-Aware Decision Support Model:

First, the integration of a sequential, two-phase algorithmic architecture successfully resolves the mathematical compensation flaw inherent in traditional additive recommendation systems. By decoupling the evaluation of non-negotiable hard constraints from the optimization of subjective soft preferences, the model ensures a mathematically sound decision-making process. The study concludes that algorithmic routing in academic advising cannot rely on linear scoring alone; it must be prefaced by strict, rule-based state validation to maintain procedural integrity.

Second, the implementation of the Boolean gatekeeper establishes a highly effective, zero-tolerance computational boundary. The finding that the system achieved a 100% rejection rate for unqualified profiles proves that strict logical AND intersections are an optimal computational method for enforcing institutional prerequisites. The study concludes that processing variables such as General Weighted Average (GWA), tuition budgets, and track alignments as binary intersection properties completely eliminates the risk of a high vocational interest score erroneously masking a fundamental academic or financial deficiency.

Third, the deployment of the Simple Additive Weighting (SAW) algorithm within a pre-filtered dataset yields highly consistent and deterministic optimization. Because the SAW equations were executed only on mathematically feasible alternatives, the resulting preference scores (V_i) were free from the distorting effects of unqualified outliers. The study concludes that when SAW normalization and weighted summation are protected by a prior constraint-resolution

phase, they provide a highly precise and computationally stable mechanism for ranking multi-dimensional O*NET vocational data.

Finally, the successful translation of mathematical weights into natural language strings confirms that algorithmic transparency can be achieved without relying on opaque heuristics. The findings demonstrate that by mapping the highest contributing variables directly to predefined templates, the system can autonomously justify its own mathematical outputs. The study concludes that the integration of an Explainable AI (XAI) module significantly elevates the utility of decision support systems, transforming them from "black-box" calculators into transparent, interpretable, and ethically accountable computational tools.

5.3 Recommendations

Drawing directly from the conclusions regarding the successful execution of the Constraint-Aware Decision Support Model, the following recommendations are proposed to further enhance the computational logic, optimize the system deployment, and expand the problem domain in future Computer Science research.

Recommendations for the Computational Model. While the two-phase algorithmic pipeline successfully prevented mathematical compensation, the Phase 2 Simple Additive Weighting (SAW) algorithm currently utilizes static, predefined weights for the O*NET soft constraints. It is recommended that future iterations of this model integrate an interactive weighting mechanism, such as the Analytic Hierarchy Process (AHP). By hybridizing the model into a Boolean-AHP-SAW architecture, the system could allow users to mathematically define the relative importance of specific vocational categories (e.g., prioritizing "Investigative" heavily over "Realistic") prior to normalization. This would dynamically adjust the w_j variables in the

summation formula, providing a more granular and highly personalized mathematical optimization without compromising the integrity of the Phase 1 hard constraint gatekeeper.

Recommendations for the System Implementation. The current Progressive Web Application (PWA) and Firebase architecture effectively demonstrated low-latency, client-side execution. However, the accuracy of the Boolean gatekeeper relies heavily on the temporal validity of the institutional datasets (e.g., shifting tuition fees or updating GWA cutoffs). It is recommended that the system implementation be upgraded to include an automated Application Programming Interface (API) bridge. Developing secure, read-only RESTful API endpoints that connect the system's NoSQL database directly to participating university registrar systems would allow the computational engine to fetch real-time institutional rules. This would eliminate the need for manual database administration and ensure that the algorithmic filtering operates on absolute, real-time academic parameters.

Recommendations for Future BSCS Research. The successful deployment of the Explainable AI (XAI) module establishes a foundation for algorithmic transparency, but its current iteration relies on static, rule-based text templates. It is highly recommended that future Bachelor of Science in Computer Science (BSCS) research explores the integration of advanced Natural Language Generation (NLG) models or localized Large Language Models (LLMs) into the XAI pipeline. Future researchers could engineer a system where an LLM is mathematically grounded by the arrays generated in Phase 2, prompting it to dynamically generate highly contextual, conversational advising summaries based strictly on the deterministic SAW outputs. Furthermore, researchers are encouraged to apply this "Constraint-Aware" algorithmic architecture to adjacent routing problems, such as automated scholarship distribution or

collegiate cross-enrollment validation, where preventing mathematical compensation remains a critical computational requirement.

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